This is an updated SRS. Changes made:

- Introduced Head Admin functionality (a mechanism for approving admin account creation requests)
- Gave the nonfunctional requirements IDs and owners
- Added a use case diagram

Jackfruit Phase-1

Software Requirements Specification

for

Online Internship & Placement Preparation Portal

Version 1.0

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Revision History

Name	Date	Reason for Changes	Version
Sharvanee, Samyuktha, Sanand, Shreya	Sep 2, 2026	Initial SRS draft	1.0

1. Introduction

1.1 Purpose This document specifies the complete software requirements for Version 1.0 of the Online Internship & Placement Preparation Portal. It defines all functional and non-functional requirements for the entire system, covering student/admin account management, internship listing and application handling, placement test practice, and performance-based material recommendation. This SRS covers the full system; no subsystem is excluded.

1.2 Intended Audience This document is intended for the development team (Samyuktha, Shreya, Sharvanee, Sanand) and the course faculty/evaluators reviewing the project. Readers new to the project should start with Section 2 for a high-level overview, then read Section 5 for detailed feature requirements relevant to their assigned module, and Section 6 for quality expectations.

1.3 Product Scope The Online Internship & Placement Preparation Portal is a web-based platform built for university engineering placement cells. It replaces manual, spreadsheet- and email-based coordination between placement officers and students with a centralized system for posting internships, managing applications, and preparing students through domain-based testing and performance analysis. The product's goal is to reduce administrative overhead for placement officers and improve student readiness by tailoring practice material to individual weak areas identified through test performance.

1.4 References

● IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications (template basis for this document)

● Project proposal document, Group 15, "Online Internship & Placement Preparation Portal"

2. Overall Description

2.1 Product Perspective This is a new, self-contained web application. It is not a follow-on to an existing system. It consists of a single web application with two role-based views (Student and Admin) sharing one backend and database, so there are no external subsystem interfaces to define beyond standard client-server communication.

2.2 Product Functions

● Account registration and profile management for students and admins

● Internship listing creation, editing, and deletion (admin) and browsing/filtering (student and admin)

● Internship application submission with eligibility validation

● Application status tracking (student) and application review/filtering (admin)

● Question and test bank management (admin)

● Test attempt, submission, and automatic scoring (student)

● Performance analysis (strengths/weaknesses) and progress tracking

● Personalized preparation material recommendation based on weak areas

2.3 User Classes and Characteristics

● Student: Engineering undergraduate seeking internships. Uses the system frequently to browse internships, apply, take tests, and check recommendations. Basic-to-moderate technical proficiency assumed; no admin privileges.

● Admin (Placement Officer): Manages internship postings, reviews applications, and curates the question bank. Uses the system regularly but performs fewer, higher-privilege actions. Requires authenticated elevated access.

● Head Admin: The same as a regular admin but also has the privilege of approval of admin account creation requests.

2.4 Operating Environment

● Client: Modern web browsers (Chrome, Firefox, Edge, Safari — latest two versions), responsive design for desktop.

● Server: Cloud computing provider, ex. AWS or Vercel.

● Database: MySQL (relational, given structured entities like students, internships, applications, questions, results).

● Frontend & backend: Next.js single-page application.

2.5 Design and Implementation Constraints

● Resume uploads restricted to PDF format, max 5 MB.

● Role-based access control (RBAC) enforced at the API layer by making students and admins hit different endpoints.

● All passwords stored using hashed encryption (e.g., bcrypt); no plaintext storage.

2.6 Assumptions and Dependencies

● Assumes users have consistent internet access while using the portal.

● Assumes the university provides accurate academic records for eligibility validation (self-reported by students at registration).

● Depends on availability of chosen hosting provider and MySQL database service.

● Assumes no integration with external Applicant Tracking Systems (ATS) or company portals in v1.0.

3. External Interface Requirements

3.1 User Interfaces

● Login/Registration screen: registration and login, form fields, and validation messages. Administrator accounts shall be created or managed only through an authorized administrative process via a head admin.

● Student Dashboard: profile summary, internship listings (with filters), application status tracker, test/practice section, recommendations panel.

● Admin Dashboard: internship management panel (add/edit/delete), applicant review panel, question bank management panel, student profile search.

● Admin Account Approval Dashboard: Visible only to head admins, this page will show all requests for admin account creation as well as the option to accept or reject them.

● Common UI elements across all screens: top navigation bar, search/filter bar wherever listings appear, consistent color-coded status tags (e.g., green = Selected, yellow = Pending, red = Rejected), standardized error/success toast messages.

3.2 Software Interfaces

● Database: MySQL 8.x — stores Users, Students, Admins, Internships, Applications, Questions, Tests, Results, and Recommendations tables.

● Full-stack Framework: Next.js, having both the frontend code as well as the backend API configuration.

● No third-party software integrations (e.g., resume parsers, external APIs) are included in v1.0.

3.3 Communications Interfaces

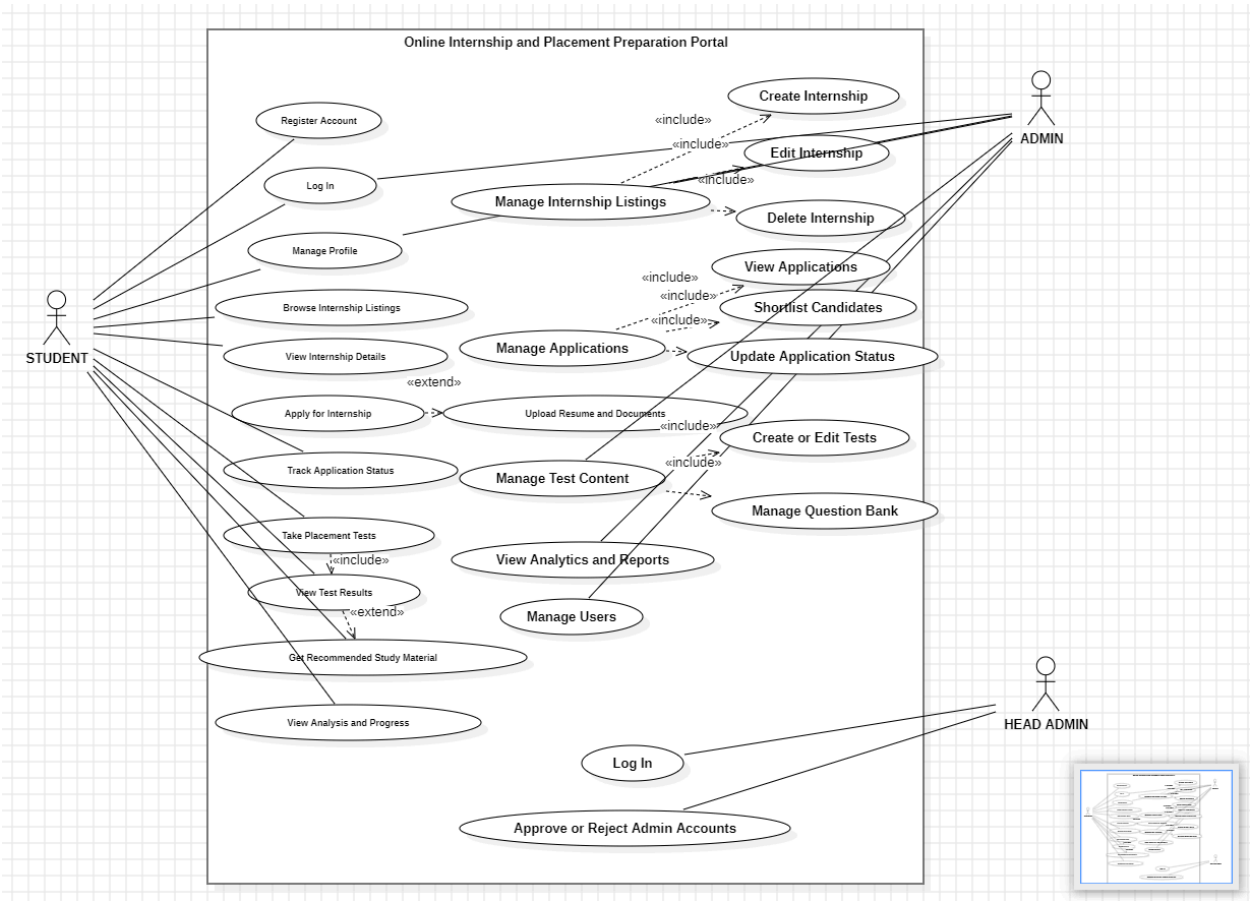
● All client-server communication occurs over HTTPS.

● REST API using JSON as the data interchange format.

● Optional email notifications (via SMTP, e.g., using Nodemailer) for application status changes. To be implemented if time permits; not mandatory for v1.0.

4. Analysis Models

Use Case Diagram :



Actors: Student, Admin, Head Admin.

● Student use cases: Register/Login, Manage Profile, Browse Internships, Apply to Internship, Track Application Status, Attempt Test, View Results, View Recommendations.

● Admin use cases: Login, Manage Internship Listings, Review Applications, Manage Question Bank, Search/Filter Student Profiles.

● Head Admin use cases: Everything a regular admin does but includes approval or rejection of admin account creation requests.

5. System Features

5.1 Account & Internship Listing Management (Owner: Samyuktha Ramesh Babu)

5.1.1 Description and Priority: This feature covers account creation and profile maintenance for both students and admins, as well as posting and browsing internship listings. Priority: High.

5.1.2 Stimulus/Response Sequences: When a student registers, they provide their personal and academic details and can optionally upload a resume before the system creates their student account. Once logged in, a student can access their profile, update their skills or academic information, and save the changes. Administrators can log in through an authorized administrative account and search or filter student profiles to view relevant details. Separately, an administrator can add a new internship by entering the company, role, domain, location, required skills, eligibility criteria, and deadline, after which the internship becomes visible to students. Students and administrators can both

apply filters on the internship listings page to narrow down results by domain, location, skills, or deadline.

5.1.3 Functional Requirements:

● REQ-1: The system shall allow students to create an account by providing their name, email, password, and academic details (branch, CGPA, and year), along with an optional resume upload in PDF format (maximum 5 MB). The system shall reject registration with a clear error message if mandatory fields are missing or the email is already registered.

● REQ-2: On registration, the system shall allow admin or head admin account creation by providing name, email, and password. Once submitted, the person may log in and view the status of their request (pending, accepted, rejected). Once the result of the request has come in (accepted, rejected), on acknowledgement (ex. pressing an "OK" button), the account may be (1) deleted on rejection (credentials will become invalid), or (2) turned into a regular admin account. The system shall reject registration with a clear error message if mandatory fields are missing or the email is already registered.

● REQ-3: Students and admins can delete their own accounts, but at any given point, there must be at least one admin account designated as a head admin.

● REQ-4: The system shall allow head admins to view all admin account creation requests on a single page, and accept or reject them.

● REQ-5: The system shall allow students to view and update their profile, skills list, and academic information, and shall allow admins to search and filter student profiles by name, branch, CGPA range, and skills.

● REQ-6: The system shall allow admins to create, update, and delete internship listings with company, role, domain, location, required skills, eligibility criteria, and application deadline, while students shall only be able to view listing details.

● REQ-7: The system shall allow both students and admins to filter internship listings by company, role, domain, location, skills, and deadline.

● REQ-8: The system shall automatically mark internship listings as “closed” when the application deadline passes.

5.2 Internship Application Management (Owner: Sanand Kumar Shetty)

5.2.1 Description and Priority: This feature lets students apply to internships and track their application status, while giving admins the ability to view applications. Priority: High.

5.2.2 Stimulus/Response Sequences: When a student clicks "Apply" on a listing, the system checks whether they meet the eligibility criteria. If they're eligible, the application gets submitted with status "Pending"; if not, the system shows a message explaining why and blocks the submission. On the admin side, an admin can open the Applications panel, filter by internship, student, or status, and update the status of any application after reviewing it.

5.2.3 Functional Requirements:

● REQ-9: The system shall allow students to submit an application for any internship listing they are eligible for.

● REQ-10: The system shall validate a student's eligibility (CGPA, skills, branch) against the internship's criteria before allowing submission, and shall block ineligible applications with an explanatory message.

● REQ-11: The system shall allow students to view the real-time status (Pending/Shortlisted/Selected/Rejected) and deadline for each of their submitted applications.

● REQ-12: The system shall allow admins to view and filter all student applications by internship, student name, or status.

● REQ-13: The system shall allow admins to update the status of any submitted application.

5.3 Placement Test & Practice (Owner: Shreya Narayana Bhat)

5.3.1 Description and Priority: This feature provides a domain-tagged question bank along with a test-taking interface for students. Priority: High.

5.3.2 Stimulus/Response Sequences: When an admin adds a new question or test, they tag it with a domain and difficulty level, after which it becomes available in the bank. A student can select a domain, usually linked to an internship they're interested in, attempt the corresponding test, and submit it, at which point the system scores it immediately.z

5.3.3 Functional Requirements:

● REQ-14: The system shall allow admins to add, edit, and remove individual questions and grouped tests, each tagged with a domain and difficulty level.

● REQ-15:The system shall allow admins to view, add, edit, delete, and filter questions and tests by domain and difficulty. Students shall only be able to view and answer questions presented to them as part of an assigned or selected test

● REQ-16: The system shall allow students to select a domain for their selected or interested internship and attempt tests relevant to the selected domain.

● REQ-17: The system shall automatically calculate and display the student's score and percentage immediately upon test submission.

● REQ-18: The system shall store each test result, including score, percentage, timestamp, and question-wise correctness, for later analysis, and shall allow students to retake practice tests while maintaining a record of previous attempts.

5.4 Result Analysis & Material Recommendation (Owner: Sharvanee Naru)

5.4.1 Description and Priority: This feature analyzes a student's test performance to identify their strengths and weaknesses and recommends suitable preparation material. Recommendations are generated using predefined rules based on the student's test performance.Priority: Medium-High.

5.4.2 Stimulus/Response Sequences: Once a student completes a test, the system updates their domain-wise performance statistics in the background. When the student opens their "My Progress" page, they can see a breakdown of their strong and weak domains along with recommended resources for improvement.

5.4.3 Functional Requirements:

● REQ-19: The system shall analyze a student's test results across attempts to identify their domains of strength and weakness, based on average performance per domain.

● REQ-20: The system shall maintain and display a student's progress trend over time, such as their score history per domain.

● REQ-21: The system shall recommend specific preparation materials targeted at each identified weak domain.

● REQ-22: The system shall automatically update a student's recommendations whenever a new test result is recorded.

● REQ-23: The system shall allow students to view their complete analysis, including strengths, weaknesses, progress, and recommendations, on a dedicated dashboard page.

6. Other Nonfunctional Requirements

6.1 Performance Requirements

● Page load time shall not exceed 4 seconds under normal network conditions.

● Test score calculation and display shall occur within 3 seconds of submission.

● Filter/search operations on listings and question banks shall return results within 3 seconds.

6.2 Safety Requirements Not physically safety-critical.

6.3 Security Requirements

● All passwords shall be stored using a one-way hash (bcrypt).

● Role-based access control shall restrict admin-only actions (listing/question management, application status updates) from student accounts.

● Students shall only be able to access and modify their own profile, applications, and results, not those of other students.

● All communication shall use HTTPS/TLS encryption.

6.4 Software Quality Attributes

● Usability: Dashboards shall be intuitive enough for a first-time user to apply to an internship or attempt a test without external guidance.

● Reliability: Test scoring and eligibility checks shall be accurate 100% of the time given valid input data.

● Maintainability: Codebase shall be modular, with each of the four feature areas (profile & listings, applications, tests, analysis & recommendation) in separate backend modules matching feature ownership.

6.5 Business Rules

● Only admins may create, edit, or delete internship listings and question bank entries.

● Students may apply to an internship only if they meet its stated eligibility criteria.

● Only the student who owns a result/recommendation record may view it; admins may view aggregate/individual student profiles but not raw test-attempt answers unless explicitly permitted.

● Domain requirement: internship domains and question/test domains shall use a shared, predefined domain list (e.g., Web Development, AI/ML, Core CS, Networks, Databases) to ensure consistent filtering and recommendation matching.

7. Other Requirements No additional legal, internationalization, or reuse requirements apply to this project at this stage.

Appendix A: Glossary

● Admin: A placement officer with elevated privileges to manage internships, applications, and the question bank.

● Head admin: An admin with the privilege to approve admin and head admin account creation requests.

● Domain: A subject/skill area (e.g., Web Development, AI/ML) used to categorize internships, questions, and tests.

● Eligibility criteria: Minimum requirements (CGPA, branch, skills) a student must meet to apply for a specific internship.

● RBAC: Role-Based Access Control — restricting system functions based on user role (Student/Admin).

Appendix B: Field Layouts

All text fields only allow ASCII characters.

Student Registration Fields: Name (text), Email (text, unique), Password (text, hashed), Branch (dropdown), CGPA (decimal), Year (dropdown), Skills (multi-select tags), Resume (PDF upload, optional, must be no more than 5MB in size).

Admin Registration Fields: Name (text), Email (text, unique), Password (text, hashed).

Internship Listing Fields: Company (text), Role (text), Domain (dropdown), Location (text), Skills Required (multi-select tags), Eligibility Criteria (min CGPA, branch), Deadline (date).

Question Fields: Question Text (text), Domain (dropdown), Difficulty (Easy/Medium/Hard), Options (4 text fields), Correct Answer (dropdown).

Report Requirements (Admin export): Student Name, Application Count, Selected Internships, Average Test Score, Weak Domains.

Appendix C: Requirement Traceability Matrix

ID	Requirement	Source	Owner	Design Reference	Test Case
REQ-1	System shall allow students to create an account with name, email, password, academic details (branch, CGPA, year), and optional resume upload (PDF, max 5MB); rejects registration on missing fields or duplicate email	5.1.3	Samyuktha	1. Go to URL of site 2. Click create student account 3. Enter name, email, password, academic details, and optionally resume	TC-01

ID	Requirement	Source	Owner	Design Reference	Test Case
				4. Click create	
REQ-2	System shall allow admin/head admin account creation via name, email, password; applicant can track request status (pending/accepted/rejected); on acknowledgement, account is deleted (rejected) or converted to admin (accepted)	5.1.3	Samyuktha	1. Go to URL of site 2. Click create admin account 3. Enter name, email, and password 4. Click create	TC-02
REQ-3	Students and admins can delete their own accounts, but at least one head admin must always remain	5.1.3	Samyuktha	1. Go to URL of site 2. Log in as student/admin 3. Navigate to account settings 4. Attempt to delete the account	TC-03
REQ-4	System shall allow head admins to view all admin account creation requests on one page and accept/reject them	5.1.3	Samyuktha	1. Go to URL of site 2. Log in as head admin 3. Navigate to requests page 4. View, accept, or reject a request	TC-04
REQ-5	Students can view/update profile, skills, academic info; admins can search/filter student profiles by name, branch, CGPA range, skills	5.1.3	Samyuktha	1. Go to URL of site 2. Log in as student/admin	TC-05

ID	Requirement	Source	Owner	Design Reference	Test Case
				3. Navigate to profile or student search page 4. Update profile or search/filter	
REQ-6	Admins can create, update, delete internship listings (company, role, domain, location, skills, eligibility, deadline); students can only view	5.1.3	Samyuktha	1. Go to URL of site 2. Log in as admin/head admin 3. Navigate to internship listing page 4. Create, update, or delete a listing	TC-06
REQ-7	Students and admins can filter internship listings by company, role, domain, location, skills, deadline	5.1.3	Samyuktha	1. Go to URL of site 2. Log in as any role 3. Navigate to listing page 4. Select filter criteria 5. View results	TC-07
REQ-8	System shall automatically mark listings "closed" once the application deadline passes	5.1.3	Samyuktha	1. Go to URL of site 2. Navigate to listing page 3. Open a listing past deadline 4. Check if applications are closed	TC-08

ID	Requirement	Source	Owner	Design Reference	Test Case
REQ-9	System shall allow students to submit an application for any internship they're eligible for	5.2.3	Sanand	1. Go to URL of site 2. Log in as eligible student 3. Select an open internship 4. Click "Apply"	TC-09
REQ-10	System shall validate eligibility (CGPA, skills, branch) before allowing submission; blocks ineligible applications with explanation	5.2.3	Sanand	1. Go to URL of site 2. Log in as student 3. Select internship 4. Attempt to apply with different CGPA/branch/skills combos	TC-10
REQ-11	Students can view real-time status (Pending/Shortlisted/Selected/Rejected) and deadline for each application	5.2.3	Sanand	1. Go to URL of site 2. Log in as student 3. Navigate to application status page 4. View applications, statuses, deadlines	TC-11
REQ-12	Admins can view and filter all applications by internship, student name, or status	5.2.3	Sanand	1. Go to URL of site 2. Log in as admin 3. Navigate to applications page	TC-12

ID	Requirement	Source	Owner	Design Reference	Test Case
				4. View/filter by internship, student, status	
REQ-13	Admins can update the status of any submitted application	5.2.3	Sanand	1. Go to URL of site 2. Log in as admin 3. Navigate to applications page 4. Select an application 5. Change status	TC-13
REQ-14	Admins can add, edit, remove individual questions and grouped tests, each tagged with domain and difficulty	5.3.3	Shreya	1. Go to URL of site 2. Log in as admin 3. Navigate to question/test bank 4. Add/edit/delete questions/tests 5. Specify domain and difficulty	TC-14
REQ-15	Admins can view/add/edit/delete/filter questions and tests by domain and difficulty; students can only view/answer assigned questions	5.3.3	Shreya	1. Go to URL of site 2. Log in as admin/student 3. Navigate to question/test section 4. Admin manages; student answers assigned test	TC-15

ID	Requirement	Source	Owner	Design Reference	Test Case
REQ-16	Students can select a domain (usually linked to an internship) and attempt relevant tests	5.3.3	Shreya	1. Go to URL of site 2. Log in as student 3. Select a domain 4. Select and attempt a relevant test 5. Submit	TC-16
REQ-17	System shall automatically calculate and display score and percentage immediately upon submission	5.3.3	Shreya	1. Go to URL of site 2. Log in as student 3. Attempt test with known correct/incorrect answers 4. Submit 5. View score/percentage	TC-17
REQ-18	System shall store each result (score, percentage, timestamp, question-wise correctness) and allow retakes while preserving history	5.3.3	Shreya	1. Go to URL of site 2. Complete a test 3. View stored result 4. Retake same test 5. Verify previous attempts remain recorded	TC-18
REQ-19	System shall analyze test results across attempts to identify	5.4.3	Sharvanee	1. Go to URL of site	TC-19

ID	Requirement	Source	Owner	Design Reference	Test Case
	strength/weakness domains based on average performance			2. Complete tests across multiple domains 3. Navigate to analysis page 4. View strengths/weaknesses	
REQ-20	System shall maintain and display progress trend over time (score history per domain)	5.4.3	Sharvanee	1. Go to URL of site 2. Complete multiple tests over time 3. Navigate to progress page 4. View score history/trends	TC-20
REQ-21	System shall recommend specific preparation materials targeted at each identified weak domain	5.4.3	Sharvanee	1. Go to URL of site 2. Complete tests resulting in weak domains 3. Navigate to recommendations page 4. View recommended materials	TC-21
REQ-22	System shall automatically update recommendations whenever a new test result is recorded	5.4.3	Sharvanee	1. Go to URL of site 2. Record existing test result 3. Complete another test	TC-22

ID	Requirement	Source	Owner	Design Reference	Test Case
				4. View recommendations again	
REQ-23	Students can view complete analysis (strengths, weaknesses, progress, recommendations) on a dedicated dashboard	5.4.3	Sharvanee	1. Go to URL of site 2. Log in as student 3. Navigate to analysis/recommendation dashboard 4. View all components	TC-23

Non-Functional Requirements

ID	Requirement	Source	Owner	Design Reference	Test Case
NFR-1	Page load time shall not exceed 4 seconds under normal network conditions	6.1 Performance	Samyuktha	1. Load any dashboard/listing page 2. Measure time-to-interactive under normal network	TC-24
NFR-2	Test score calculation and display shall occur within 3 seconds of submission	6.1 Performance	Shreya	1. Submit a completed test 2. Measure time from submission to score display	TC-25
NFR-3	Filter/search on internship listings shall return results within 3 seconds	6.1 Performance	Samyuktha	1. Apply a filter on the listings page 2. Measure response time	TC-26

ID	Requirement	Source	Owner	Design Reference	Test Case
NFR-4	Filter/search on the question bank shall return results within 3 seconds	6.1 Performance	Shreya	1. Apply a filter in the question/test bank 2. Measure response time	TC-27
NFR-5	All passwords shall be stored using a one-way hash (bcrypt)	6.3 Security	Samyuktha	1. Register a new account 2. Inspect stored credential in the database	TC-28
NFR-6	RBAC shall restrict admin-only actions (listing/question management, status updates) from student accounts	6.3 Security	Samyuktha	1. Log in as student 2. Attempt admin-only endpoint/action	TC-29
NFR-7	Students shall only access/modify their own profile, applications, and results	6.3 Security	Sanand	1. Log in as Student A 2. Attempt to view/edit Student B's data	TC-30
NFR-8	All communication shall use HTTPS/TLS encryption	6.3 Security	Samyuktha	1. Inspect network requests 2. Verify TLS on every endpoint	TC-31
NFR-9	Usability: dashboards intuitive enough for a first-time user to apply/attempt a test without guidance	6.4 Quality	Samyuktha	1. Give new user task with no instructions 2. Observe unaided completion	TC-32
NFR-10	Reliability: test scoring shall be accurate 100% of the time given valid input	6.4 Quality	Shreya	1. Submit test with known correct/incorrect answers 2. Verify score matches exactly	TC-33

ID	Requirement	Source	Owner	Design Reference	Test Case
NFR-11	Reliability: eligibility checks shall be accurate 100% of the time given valid input	6.4 Quality	Sanand	1. Apply with known-eligible/ineligible profiles 2. Verify outcome correctness	TC-34
NFR-12	Maintainability: codebase shall be modular, matching the four feature-ownership areas	6.4 Quality	Shared (all)	1. Review repo structure 2. Confirm module boundaries match ownership	TC-35
NFR-13	Only admins may create/edit/delete internship listings and question bank entries	6.5 Business Rules	Samyuktha / Shreya	1. Log in as student 2. Attempt listing/question CRUD	TC-36
NFR-14	Students may apply to an internship only if they meet its stated eligibility criteria	6.5 Business Rules	Sanand	Cross-referenced with REQ-10 eligibility validation	TC-10
NFR-15	Only the owning student may view their result/recommendation record; admins see aggregates, not raw answers	6.5 Business Rules	Sharvanee	1. Log in as Student A 2. Attempt to view Student B's results/recommendations	TC-37
NFR-16	Internship and question/test domains shall use a shared, predefined domain list	6.5 Business Rules	Samyuktha	1. Create internship and question with domain field 2. Verify same dropdown list used	TC-38

Appendix D: Test case enumeration

Here's the full Appendix D table from the PDF, including the original TC-01 to TC-23 and the new TC-24 to TC-38 added for non-functional requirements.

ID	Purpose	Input	Pass Output	Fail Output
TC-01	Checking student account creation with valid and invalid information	Valid student registration details, followed by missing/invalid mandatory fields and duplicate email. Optional resume may be provided	Valid registration creates the account; invalid or duplicate registration is rejected with a clear error message	Invalid registration is accepted, or valid registration is rejected
TC-02	Checking admin account creation and request submission	Valid admin name, email, and password, followed by invalid/missing information or duplicate email	Valid request is submitted with pending status; invalid/duplicate registration is rejected with a clear error message	Invalid request is accepted or valid request is rejected
TC-03	Checking account deletion while ensuring at least one head admin remains	Student/admin attempts to delete their own account; only remaining head admin attempts deletion	Student/admin account can be deleted; deletion of the only head admin is blocked	Account deletion violates the minimum head admin requirement or valid deletion is blocked
TC-04	Checking head admin management of admin account creation requests	Head admin views pending requests, accepts one, rejects another	Requests displayed correctly; accepted account becomes regular admin after acknowledgement; rejected account becomes invalid/removed	Requests cannot be managed correctly, or accepted/rejected accounts have the wrong final state

ID	Purpose	Input	Pass Output	Fail Output
TC-05	Checking student profile updates and admin profile search/filtering	Student changes valid/invalid profile/academic/skills info; admin searches by name, branch, CGPA range, skills	Valid changes saved, invalid changes rejected; admin receives correct matching profiles	Invalid profile info is saved or search/filter results are incorrect
TC-06	Checking internship listing create, update, delete	Admin creates an internship, modifies it, deletes it	Listing created, updated, deleted successfully with correct info displayed	One or more operations fails or produces incorrect listing info
TC-07	Checking internship listing filtering	Company, role, domain, location, skills, and/or deadline filter criteria	Only matching internships displayed	Non-matching internships displayed or matches omitted incorrectly
TC-08	Checking automatic closure of internship after deadline	Internship whose deadline has passed	Internship auto-marked closed; students cannot apply	Internship remains open or accepts applications after deadline
TC-09	Checking application submission for an eligible student	Logged-in eligible student selects an open internship	Application submitted successfully with status "Pending"	Eligible student cannot submit, or application gets wrong initial status
TC-10	Checking internship eligibility validation	Students with different CGPA, branch, skills combos attempt to apply	Application allowed only when all criteria satisfied; otherwise blocked with explanation	Ineligible student can submit, or eligible student is incorrectly blocked

ID	Purpose	Input	Pass Output	Fail Output
TC-11	Checking student application status and deadline tracking	Student has applications with different statuses and deadlines	Student can view current status and deadline for every application	Status/deadline info missing, outdated, or incorrect
TC-12	Checking admin application viewing and filtering	Admin filters applications by internship, student name, status	Relevant applications displayed; filters return correct results	Applications missing or filtering returns incorrect results
TC-13	Checking application status updates by admins	Admin changes status between Pending, Shortlisted, Selected, Rejected	New status saved and correctly reflected in student's tracker	Status not saved or incorrect status displayed
TC-14	Checking question and test create/update/delete	Admin adds, edits, deletes questions/tests with domain and difficulty	Questions/tests created, updated, deleted with correct info	One or more operations fails or incorrect info stored
TC-15	Checking question/test viewing, filtering, role-based access	Admin filters by domain/difficulty; student attempts management functionality	Admin can view/filter; student can only access assigned test questions, not manage bank	Student can modify/manage questions, or admin filtering fails
TC-16	Checking student selection/attempt of domain-specific test	Student selects domain and associated test, answers, submits	Relevant test presented and completed/submitted successfully	Student receives unrelated test or cannot complete/submit

ID	Purpose	Input	Pass Output	Fail Output
TC-17	Checking automatic score/percentage calculation	Completed test with known correct/incorrect answers	System immediately calculates and displays correct score/percentage	Score/percentage incorrect, missing, or not displayed
TC-18	Checking test result storage and retake functionality	Student completes/submits test, views result, retakes same test	Score, percentage, timestamp, question-wise correctness stored per attempt; previous attempts remain	Result info not stored correctly, or previous attempts overwritten/lost
TC-19	Checking identification of strengths and weaknesses	Test results with different average performance across domains	System correctly identifies strong/weak domains based on average performance	Analysis doesn't correspond to stored performance
TC-20	Checking display of student progress trends	Multiple test attempts across different dates/domains	Score history/progress trend displayed correctly per domain	Progress history missing or doesn't reflect results
TC-21	Checking recommendation of prep material for weak domains	Student has identified weak domain(s)	Materials relevant to each weak domain recommended	Recommendations missing or unrelated to weak domains
TC-22	Checking automatic update of	Student completes new test changing	Recommendations auto-updated	Recommendations unchanged or based on outdated results

ID	Purpose	Input	Pass Output	Fail Output
	recommendations after new result	domain-wise performance	based on new result	
TC-23	Checking complete analysis and recommendation dashboard	Student with recorded results opens dashboard	Dashboard displays strengths, weaknesses, progress, recommendations correctly	One or more required components missing or incorrect
TC-24	Checking page load performance across dashboards/listings (Owner: Samyuktha)	Any dashboard or listing page under normal network conditions	Page becomes interactive within 4 seconds	Page load exceeds 4 seconds
TC-25	Checking test score display timing (Owner: Shreya)	A completed test is submitted	Score and percentage displayed within 3 seconds of submission	Score display takes longer than 3 seconds or doesn't appear
TC-26	Checking internship listing filter/search response time (Owner: Samyuktha)	A filter is applied on the listings page	Filtered results returned within 3 seconds	Results take longer than 3 seconds or fail to return
TC-27	Checking question bank filter/search response time (Owner: Shreya)	A filter is applied in the question/test bank	Filtered results returned within 3 seconds	Results take longer than 3 seconds or fail to return
TC-28	Checking password hashing (Owner: Samyuktha)	A new account is registered	Stored credential is a bcrypt hash, never plaintext	Password stored in plaintext or with reversible/weak hash

ID	Purpose	Input	Pass Output	Fail Output
TC-29	Checking RBAC enforcement against student accounts (Owner: Samyuktha)	Logged-in student attempts an admin-only action	Action blocked with an appropriate authorization error	Student is able to perform the admin-only action
TC-30	Checking student data access isolation (Owner: Sanand)	Student A attempts to view/edit Student B's profile, applications, or results	Access denied for all of Student B's private data	Student A can view or modify Student B's data
TC-31	Checking HTTPS/TLS on all communication (Owner: Samyuktha)	Network traffic inspected across all major flows	All requests use HTTPS/TLS, no plaintext endpoints	Any endpoint reachable over plain HTTP
TC-32	Checking first-time usability without guidance (Owner: Samyuktha)	First-time user asked to apply/attempt a test with no instructions	User completes the task unaided	User cannot complete the task without help
TC-33	Checking test scoring reliability (Owner: Shreya)	Test with known correct/incorrect answers submitted	Computed score matches expected score exactly, every time	Computed score differs from expected on any run
TC-34	Checking eligibility check reliability (Owner: Sanand)	Known-eligible/ineligible profiles attempt to apply	Eligible always allowed, ineligible always blocked	Any eligible student blocked or ineligible student allowed
TC-35	Checking modular codebase structure (Owner: Shared – all)	Repository/codebase structure	Four feature areas exist as separate backend modules matching ownership	Feature areas intermixed with unclear ownership boundaries

ID	Purpose	Input	Pass Output	Fail Output
TC-36	Checking only admins can manage listings/question bank (Owner: Samyuktha/Shreya)	Student attempts to create/edit/delete a listing or question bank entry	Action blocked for student; only admin/head admin can perform it	Student account able to perform the action
TC-37	Checking result/recommendation record privacy (Owner: Sharvanee)	Student A attempts to view Student B's results/recommendations	Access denied; only owning student can view own record	Student A can view Student B's results/recommendations
TC-38	Checking shared predefined domain list consistency (Owner: Samyuktha)	Internship listing and question bank entry created with domain fields	Both use the same predefined domain list, no divergence	Internship and question/test domain lists differ or inconsistent